

NetworkProgramming Library Project Documentation

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Overview and Purpose

This library provides a comprehensive implementation of network programming fundamentals in C/C++, featuring TCP client and server components along with implementations of standard protocols: **Daytime** (RFC 867) and **Echo** (RFC 862). The project serves as an educational resource for learning BSD socket APIs, testing TCP connections, and demonstrating the construction of simple network services.

Class Architecture

The following section provides a concise overview of the main classes. Detailed method documentation is available in the generated Doxygen documentation.

- **Socket**: Base class managing the socket file descriptor lifecycle (creation, closure).
- **ClientTCP**: TCP client implementation that creates a socket and establishes connections to remote servers via `connect()`.
- **ServerTCP**: TCP server implementation that creates, binds, and listens on a specified port. Includes utility methods for `accept()`, `logConnection()`, `connectionServiced()`, and an `onServerStart()` hook for customization.
- **ServerTCPdaytime/ClientTCPdaytime**: Implementation of the Daytime Protocol where the server transmits the current date and time to clients upon connection.
- **ServerTCPecho/ClientTCPecho**: Implementation of the Echo Protocol where the server echoes back any data received from the client.

Build and Deployment (CMake)

Prerequisites

- Linux operating system
- C++17 compliant compiler (GCC 7+ or Clang 5+)
- CMake version 3.10 or higher

- Build tools (make or ninja)
- Optional: Doxygen

Build Instructions

1. Navigate to the project directory: `NetworkProgrammingLib/`
2. Create a build directory and configure the project:

```
cmake -S . -B build
```

3. Build the project:

```
cmake --build build
```

Executables and the static library are generated in the `build` directory. Available targets include:

- `test_serverTCP`, `test_clientTCP`
- `test_serverTCPdaytime`, `test_clientTCPdaytime`
- `test_serverTCPEcho`, `test_clientTCPEcho`

Running the Applications

- **Daytime Server:**

```
./test_serverTCPdaytime [port]
```

Default port: 13 (or specify custom port, e.g., 8013)

- **Daytime Client:**

```
./test_clientTCPdaytime <server-IP> <port>
```

- **Echo Server:**

```
./test_serverTCPEcho [port]
```

Default port: 7 (or specify custom port, e.g., 8007)

- **Echo Client:**

```
./test_clientTCPEcho <server-IP> <port>
```

Enter lines of text to receive echoed responses from the server.

Note: Ports 7 and 13 are privileged ports and may require administrative privileges. For testing purposes, use unprivileged ports (1024–65535), such as 8000 or 8007.

Additional Resources

Generating API Documentation with Doxygen

The project includes comprehensive API documentation that can be generated using Doxygen:

```
cd NetworkProgramming
doxygen Doxyfile
```

Conclusion

This project demonstrates clean, well-documented implementations of TCP clients and servers for the Daytime and Echo protocols. It serves as a foundational resource for understanding network programming concepts using BSD sockets and provides a practical framework for experimenting with TCP-based communication patterns.